

fn make_expr_datetime_component

Michael Shoemate (@Shoeboxam)

This proof resides in “contrib” because it has not completed the vetting process.

Proves soundness of `make_expr_datetime_component` in `mod.rs` at commit `f5bb719` (outdated¹).
`make_expr_datetime_component` returns a Transformation that extracts a component from a temporal data type.

1 Hoare Triple

Precondition

Compiler-verified

- Argument `input_domain` of type `ExprDomain`
- Argument `input_metric` of type `M`
- Generic `M` implements `OuterMetric`
 - `OuterMetric` defines an associated type `InnerMetric` that must implement `DatasetMetric`
- `(ExprDomain, M)` implements `MetricSpace`
- `Expr` implements `StableExpr<M, M>`

Caller-verified

None

Pseudocode

```
1 def make_expr_datetime_component(  
2     input_domain: ExprDomain,  
3     input_metric: M,  
4     expr: Expr,  
5 ) -> Transformation:  
6     match expr: #  
7         case Expr.Function(input=inputs, function=FunctionExpr.TemporalExpr(  
8             temporal_function)):  
9             pass  
10            case _:  
11                raise ValueError("expected datetime component expression")  
12            to_dtype, _ = match_datetime_component(temporal_function) #  
13
```

¹See new changes with `git diff f5bb719..75a77e0 rust/src/transformations/make_stable_expr/namespace_dt/expr_datetime_component/`
`rs`

```

14 # raises an error if there is not exactly one input
15 input, = inputs #
16
17 t_prior = input.make_stable(input_domain, input_metric) #
18 middle_domain, middle_metric = t_prior.output_space()
19
20 in_dtype = middle_domain.column.dtype
21 if in_dtype not in {DataType.Time, DataType.Datetime, DataType.Date}: #
22     raise ValueError("expected a temporal input type")
23
24 output_domain = middle_domain.clone() #
25 output_domain.column.set_dtype(to_dtype) #
26
27 def function(expr: Expr) -> Expr:
28     return Expr.Function(
29         input=[expr],
30         function=FunctionExpr.TemporalExpr(temporal_function),
31         options=FunctionOptions(
32             collect_groups=ApplyOptions.ElementWise,
33         ),
34     )
35
36 return t_prior >> Transformation.new( #
37     middle_domain,
38     output_domain,
39     Function.then_expr(function),
40     middle_metric,
41     middle_metric,
42     StabilityMap.new(lambda d_in: d_in),
43 )

```

Postcondition

Theorem 1.1. For every setting of the input parameters (input_domain , input_metric , M) to `make_expr_datetime_component` such that the given preconditions hold, `make_expr_datetime_component` raises an error (at compile time or run time) or returns a valid transformation. A valid transformation has the following properties:

1. (Data-independent runtime errors). For every pair of members x and x' in input_domain , `invoke( $x$ )` and `invoke( $x'$ )` either both return the same error or neither return an error.
2. (Appropriate output domain). For every member x in input_domain , `function( $x$ )` is in output_domain or raises a data-independent runtime error.
3. (Stability guarantee). For every pair of members x and x' in input_domain and for every pair $(d_{\text{in}}, d_{\text{out}})$, where d_{in} has the associated type for input_metric and d_{out} has the associated type for output_metric , if x, x' are d_{in} -close under input_metric , `stability_map( $d_{\text{in}}$ )` does not raise an error, and `stability_map( $d_{\text{in}}$ ) =  $d_{\text{out}}$` , then `function( $x$ )`, `function( $x'$ )` are d_{out} -close under output_metric .

Starting from line 6, `expr` is matched as if it were a temporal expression, or otherwise rejects the expression. All preconditions for `make_datetime_component` on line 12 are satisfied by the compiler, therefore by the postcondition `to_dtype` and `max_num_partitions` are the output data type and upper bound on the number of unique values in the output.

All preconditions for `make_stable` on line 17 are compiler-verified, therefore by the postcondition `t_prior` is a valid transformation. To prove that the output is a valid transformation, we must first prove that the transformation on line 36 is a valid transformation.

Proof of Data-Independent Errors. If the input data type does not include the component, then all possible choices of input dataset would fail, resulting in data-independent errors.

If the input data type does include the component, then Polars will never raise an error. $\square$

Proof of Output Domain. The output domain is the same as the input domain, but the active series has a new data type and margin metadata needs to be updated. Line 25 updates the series domain so that non-null elements are of type `to_type`, which by the postcondition of `match_datetime_component` on line 12 represents the type of outputs when `temporal_function` is applied to the data.

Other domain descriptors, like the nullity and name, remain unchanged. New null values cannot be introduced, because retrieval of a time component fails (in a data-independent way) if the component is not present in the type. The output domain now accounts for all transformations made to data in the input domain. $\square$

Proof of Stability. Datetime component retrieval is a 1-stable row-by-row transformation, as the component retrieval is applied independently to each and every row. See `make_row_by_row` for a proof of the stability of row-by-row functions. $\square$

Since it has been shown that both `t_prior` and the component transformation are valid transformations, then the preconditions for `make_chain_tt` are met (invoked via the right-shift operator shorthand), and therefore the output is a valid transformation.